Solid State Physics — Week 7 Assignment Questions & Answers

Question 61

A copper wire of 1 mm diameter carries a current of$(5\ \mathrm{mA})$. Calculate the average drift velocity of the free electrons. Copper is monovalent and crystallizes in FCC structure with a lattice parameter of 3.61 Å.

(a) $3.42 \times 10^{-6}\ m/s$

(b) $4.68 \times 10^{-7}\ m/s$

(c) $5.82 \times 10^{-9}\ m/s$

(d) $6.17 \times 10^{-10}\ m/s$

Accepted answer: (b)

Question 62

In copper the number free electrons per unit volume is $8.5 \times 10^{28}\ /m^3$ and its conductivity$(\sigma = 6 \times 10^7\ \Omega^{-1}m^{-1})$. Calculate the relaxation time of electron when electronic mass and charge are $9.1 \times 10^{-31}\ kg$ and $1.6 \times 10^{-19}\ C$ respectivelly.

(a) $1.17 \times 10^{-12}s$

(b) $2.39 \times 10^{-13}\ s$

(c) $2.50 \times 10^{-14}\ s$

(d) $4.93 \times 10^{-15}\ s$

Accepted answer: (c)

Question 63

Assuming that each atom of copper contributes one free electron, calculate the drift velocity of free electrons in the copper conductor of cross-sectional area $10^{-4}\ m^2$ and carrying a current of 200 A. (Given: Atomic weight of Cu is 63.5, density of copper is $8.94 \times 10^3\ kg/m^3$, electronic charge is $1.602 \times 10^{-19}\ C$ and Avogadro's number $N_A = 6.02 \times 10^{23}$).

(a) $1.47 \times 10^{-4}\ m/s$

(b) $2.21 \times 10^{-6}\ m/s$

(c) $3.38 \times 10^{-8}\ m/s$

(d) $4.24 \times 10^{-10}\ m/s$

Accepted answer: (a)

Question 64

Estimate classically the relaxation time (τ) of electrons in the copper when its resistivity is $1.69 \times 10^{-8}\ \Omega - m$ and the concentration of free electrons in copper($n = 8.5 \times 10^{28}\ /m^3$).

(a) $1.13 \times 10^{-12} s$

(b) $2.47 \times 10^{-14}\ s$

(c) $3.17 \times 10^{-16}\ s$

(d) $4.97 \times 10^{-18}\ s$

Accepted answer: (b)

Question 65

Estimate classically the mean free path (λ) of electrons in copper at temperature 20°C. Relaxation time of electrons in copper is$(2.5 \times 10^{-14}s)$. [Given: Boltzmann constant$(K_B = 1.38 \times 10^{-23}\, J/K)$]

(a) 2.88 nm

(b) 3.33 nm

(c) 4.44 nm

(d) 5.55 nm

Accepted answer: (a)

Question 66

The wave function of electrons in a one-dimensional potential box of dimension 'a' is given by $\Psi_n = A \sin\left(\frac{n\pi}{a}\right) x$ where n = 1, 2, 3 ....

Evaluate A by normalizing the wave function to unity.

(a) $\sqrt{\frac{4}{a}}$

(b) $\sqrt{\frac{3}{a}}$

(c) $\sqrt{\frac{2}{a}}$

(d) $\sqrt{\frac{a}{2}}$

Accepted answer: (c)

Question 67

In a one-dimensional potential box of length L, the maximum wavelength that an electron can have is

(a) $\frac{L}{4}$

(b) $\frac{L}{2}$

(c) L

(d) 2L

Accepted answer: (d)

Question 68

In an one-dimensional lattice of interatomic separation a, the maximum wavelength of the electron gets Bragg reflected is

(a) $\frac{a}{4}$

(b) $\frac{a}{2}$

(c) a

(d) $2a$

Accepted answer: (d)

Question 69

Calculate the energy difference between the ground state energy and the first excited state energy for free electrons in a three dimensional potential box of dimension $1 \times 1 \times 1\ cm^3$. [Given: Planck constant$(h = 6.6 \times 10^{-34}\ J - s)$]

(a) $1.795 \times 10^{-33}\ J$

(b) $4.953 \times 10^{-35}\ J$

(c) $6.123 \times 10^{-39}\ J$

(d) $7.128 \times 10^{-45}\ J$

Accepted answer: (a)

Question 70

In a one-dimensional solid of interatomic separation 2Å, find the critical or minimum momentum values for which Bragg reflections take place. [Given: Planck constant$(h = 6.64 \times 10^{-34}\ J - s)$]

(a) $2.17 \times 10^{-12}\ kg.m/s.$

(b) $3.22 \times 10^{-16}\ kg.m/s$

(c) $5.12 \times 10^{-20}\ kg.m/s$

(d) $1.66 \times 10^{-24}\ kg.m/s$

Accepted answer: (d)